The Relationship Between Foreign Direct Investment and Economic Growth: The Role of Political Stability and Corruption

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April 29 2025

1 Introduction

Foreign Direct Investment (FDI) is widely recognized as a key driver of economic growth and development, especially in emerging markets. FDI can bring in capital, create jobs, transfer technology, and improve infrastructure, all of which contribute to long-term economic growth. However, the impact of FDI on GDP growth is not consistent across countries, suggesting that institutional factors, such as political stability and corruption, may play a crucial role in shaping this relationship.

Political stability provides a secure environment for investment and long-term business planning, while corruption can create uncertainty, increase business costs, and deter foreign investors. Corruption not only affects the willingness of foreign investors to enter a market but also influences how effectively the investment is utilized. High levels of corruption may lead to misallocation of resources, inefficiencies in public infrastructure, and increased costs for businesses, all of which could diminish the positive effects of FDI on economic growth. On the other hand, strong political stability and low corruption can create a predictable and transparent business environment, encouraging sustained foreign investment and maximizing its contribution to economic performance.

Understanding how these two governance factors interact with FDI and influence economic performance is critical for policymakers seeking to create favorable investment environments. This project aims to analyze how political stability and corruption affect the relationship between FDI and GDP growth across a diverse set of countries. Specifically, the project will explore whether political stability and corruption moderate the effect of FDI on economic growth, and whether the interaction between these factors leads to variations in macroeconomic performance.

The data for this project comes from two key sources: macroeconomic indicators such as GDP growth, FDI (% of GDP), inflation, trade balance, and unemployment are obtained from the World Bank, while annual scores for political stability and control of corruption are drawn from the World Governance Indicators (WGI). The project will involve exploratory data analysis, statistical modeling, and visualization to uncover patterns and relationships in the data. By identifying how political stability and corruption influence the effectiveness of FDI, this project will provide insights into how governance quality can enhance or weaken the benefits of foreign investment. The results are expected to provide valuable guidance for both policymakers and international investors in improving the overall investment climate and fostering sustainable economic growth.

2 Methods

This project follows a structured approach involving data acquisition, data cleaning, merging, and exploratory data analysis (EDA) to prepare the dataset for modeling and generate preliminary insights.

The data was sourced from two key sources. Macroeconomic indicators—including GDP growth, FDI (% of GDP), inflation, trade balance, and unemployment—were obtained from the World Bank through its public API using Python's requests and pandas libraries. Governance indicators, specifically political stability and control of corruption, were retrieved from the Worldwide Governance Indicators (WGI) dataset, which provides annual scores on governance quality across countries from 1996 onward.

The data wrangling process involved several steps to ensure consistency and completeness. First, mismatched country names between the World Bank and WGI datasets were corrected to ensure consistency. For instance, "Cote d'Ivoire" was renamed to "Côte d'Ivoire" and "Czechia" to "Czech Republic." Year consistency was also addressed—data for the year 2001 was missing for all countries in the WGI dataset. Forward and backward filling were applied where possible to avoid breaks in the time-series structure while maintaining data continuity.

Handling missing data required a careful approach to preserve the integrity of the panel structure. Countries with any missing values for the two primary variables of interest—GDP growth and FDI—were dropped from the dataset to avoid bias in the modeling phase. For other macroeconomic indicators, missing values were addressed using a combination of forward and backward fill within each country. Linear interpolation was applied to fill remaining gaps in unemployment and trade balance data, ensuring that the dataset remained balanced while minimizing data loss.

Extreme outliers in FDI and inflation were clipped at the 1st and 99th percentiles to reduce the influence of rare but extreme events without entirely removing data points. This approach allowed the model to focus on typical patterns in the data without being distorted by outliers.

The World Bank and WGI datasets were merged using a left join based on country and year. Country names were harmonized between the datasets, and non-matching entries were reviewed manually. This ensured that the panel structure remained intact and that no key observations were lost during the merge process.

To improve model interpretability and performance, political stability and corruption scores were converted from continuous values to five categorical levels using quantiles. The five levels were defined as "Very Low," "Low," "Moderate," "High," and "Very High." This allowed for a more structured analysis of interaction effects and facilitated better performance for tree-based models like decision trees and XGBoost, which are known to perform well with categorical data. The original continuous values were retained for use in other statistical models, including panel regression.

Exploratory Data Analysis (EDA) was conducted using Python's matplotlib, seaborn, and pandas libraries. Summary statistics were generated for all key variables, including measures of central tendency (mean, median), spread (variance, standard deviation), and distribution (skewness, kurtosis). Correlation matrices were computed to identify relationships between FDI, GDP growth, and governance indicators. Distribution plots and time-series trend lines were created to identify patterns over time and across countries. Interaction exploration was conducted by binning political stability and corruption into categorical levels and examining how these categories influenced the relationship between FDI and GDP growth.

The cleaned and merged dataset covers a balanced panel of countries from 2000 to 2023, with governance indicators and macroeconomic metrics aligned across time periods. The data is now prepared for the modeling phase, where panel regression and tree-based models will be used to test the influence of political stability and corruption on the FDI-GDP growth relationship.

In the final phase of the project, we refined our modeling strategy based on insights from exploratory analysis. We reverted to using the original continuous governance scores—political stability and corruption—instead of quantile-based levels. This decision was motivated by the observation that interaction terms involving continuous scores provided stronger and more interpretable effects in panel regression, and allowed tree-based models to discover nonlinear relationships more flexibly.

The modeling phase involved two main approaches. First, we used a fixed-effects panel regression model to test whether political stability and corruption scores moderate the relationship between FDI (% of GDP) and GDP growth. Interaction terms between FDI and each governance score were included to evaluate moderation effects. Second, we employed XGBoost (eXtreme Gradient Boosting), a gradient boosting framework that builds an ensemble of shallow decision trees to model nonlinearities and interactions. XGBoost is particularly effective for tabular data and was used here to complement the regression analysis by uncovering more complex patterns.

To interpret the XGBoost model, we used SHAP (SHapley Additive exPlanations), a unified framework for model interpretability based on cooperative game theory. SHAP assigns each feature a contribution value to each prediction, making it possible to understand both main effects and interactions. SHAP summary plots were used to identify the most influential features on GDP growth, while SHAP interaction plots revealed how the impact of FDI varied across levels of political stability and corruption.

Because our goal was interpretation rather than prediction, we trained each XGBoost model on the

entire dataset using a set of predefined hyperparameter configurations. Rather than optimizing for predictive performance, we evaluated model stability and interpretability by comparing feature importance rankings and SHAP value patterns across runs. While no formal cross-validation was conducted, the inclusion of multiple model settings helped ensure that the insights we derived were robust and not the result of a single model specification. We also reported R^2 and Root Mean Squared Error (RMSE) on the training data as basic indicators of model fit. These metrics were not used to select a final model but to ensure that the learned relationships reflected meaningful structure in the data.

3 Results

Table 1 presents the results from a fixed-effects panel regression model estimating the relationship between FDI (% of GDP) and GDP growth, with interaction terms for political stability and corruption. The coefficient for FDI is positive and statistically significant, indicating a general growth-enhancing effect. The interaction term between FDI and political stability score is negative and significant ($p < 0.05$), suggesting that the effectiveness of FDI diminishes in more politically stable countries. In contrast, the interaction between FDI and corruption score is not significant, providing little evidence that corruption modifies the FDI-growth relationship.

To further explore these relationships, we trained an XGBoost model on the full dataset to capture potential nonlinearities and complex interactions. The model achieved an R^2 of 0.406 and a Root Mean Squared Error (RMSE) of 3.031 on the training data. While not optimized for prediction, these values indicate that the model captures meaningful structure in the data and is suitable for interpretability analysis using SHAP.

Variable	**Coefficient**	**Std. Error**	**t-Statistic**	**P-value**
Intercept	6.8114	0.908	7.501	< 0.001
FDI (% of GDP)	0.1297	0.013	9.809	< 0.001
Corruption Score	-0.0021	0.345	-0.006	0.995
Political Stability Score	1.0896	0.215	5.056	< 0.001
FDI × Corruption	-0.0205	0.016	-1.283	0.199
FDI × Political Stability	-0.0497	0.022	-2.248	0.025
Inflation (%)	0.0061	0.015	0.420	0.675
Unemployment (%)	-0.1789	0.032	-5.658	< 0.001
Trade Balance (% of GDP)	0.0510	0.011	4.703	< 0.001
Model Summary (Regression Fit Statistics)				
	R^2	**Adjusted R^2**	**RMSE**	
	0.201	0.164	3.031	
	F-statistic (p)	**AIC**	**BIC**	
	5.486 ($p < 0.001$)	17500	18360	

Table 1: Results from a fixed-effects panel regression model estimating the impact of FDI, political stability, and corruption on GDP growth. Country fixed effects are included but not displayed. The interaction between FDI and political stability is statistically significant and negative, indicating that FDI has a stronger positive effect on GDP growth in politically unstable environments.

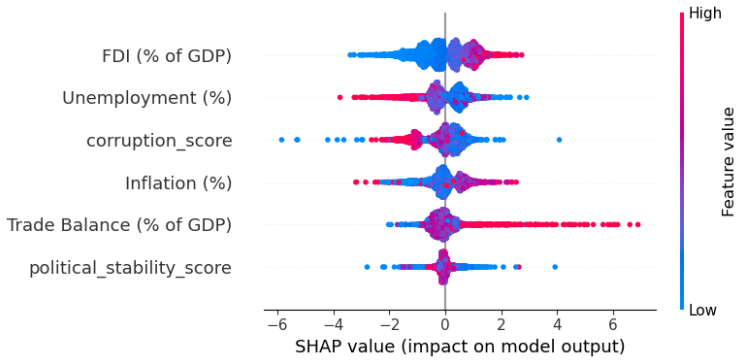

Figure 1: SHAP summary plot showing the contribution of each feature to the XGBoost model's prediction of GDP growth. FDI (% of GDP) is the most influential variable, followed by unemployment and trade balance. Governance scores have weaker direct effects.

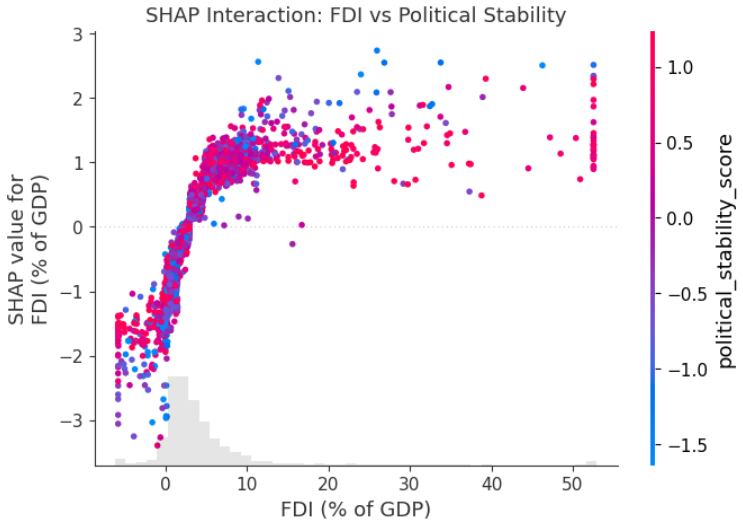

Figure 2: SHAP interaction plot for FDI and political stability. Darker blue points (lower political stability) are associated with stronger positive effects of FDI on GDP growth.

Figure 1 shows the SHAP summary plot, which ranks features by their contribution to the model's predictions of GDP growth. FDI emerges as the most influential predictor, with low values (blue) contributing negatively and high values (red) contributing positively to the output. In contrast, political stability and corruption scores have smaller and more symmetric effects around zero, aligning with the regression result that their direct effects on GDP growth are limited.

Figure 2 illustrates the SHAP interaction between FDI and political stability. The color gradient indicates that when political stability is low (blue), FDI has a stronger positive effect on GDP growth. This relationship weakens as stability increases. The plot confirms the regression finding that FDI is more impactful in politically unstable environments, potentially because it compensates for weak institutions or investment gaps.

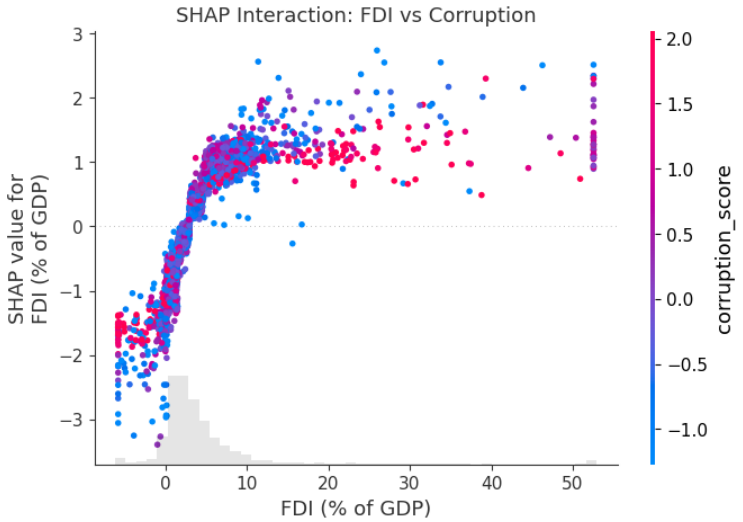

Figure 3: SHAP interaction plot for FDI and corruption. Interaction effects are weaker and less distinct than for political stability, showing a more uniform influence of FDI across corruption levels.

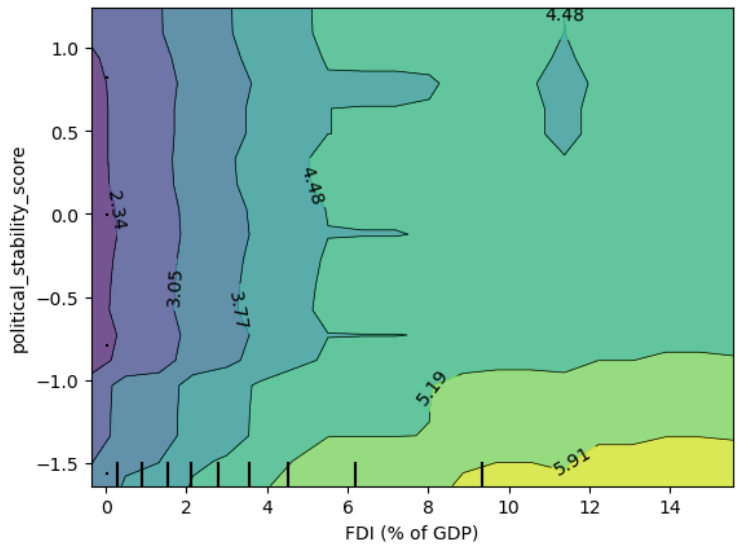

Figure 4: Partial dependence of GDP growth on FDI (% of GDP) and political stability score. The plot shows that FDI has a stronger positive marginal effect in countries with lower political stability, as indicated by steeper gradients on the left side of the vertical axis.

Figure 3 displays the SHAP interaction between FDI and corruption. Unlike political stability, the interaction here is less pronounced. While there is a weak tendency for FDI to be more beneficial in less corrupt contexts, the color patterns are more dispersed and inconsistent. This visual reinforces the regression finding that corruption does not significantly moderate the FDI-growth relationship in a consistent or strong manner.

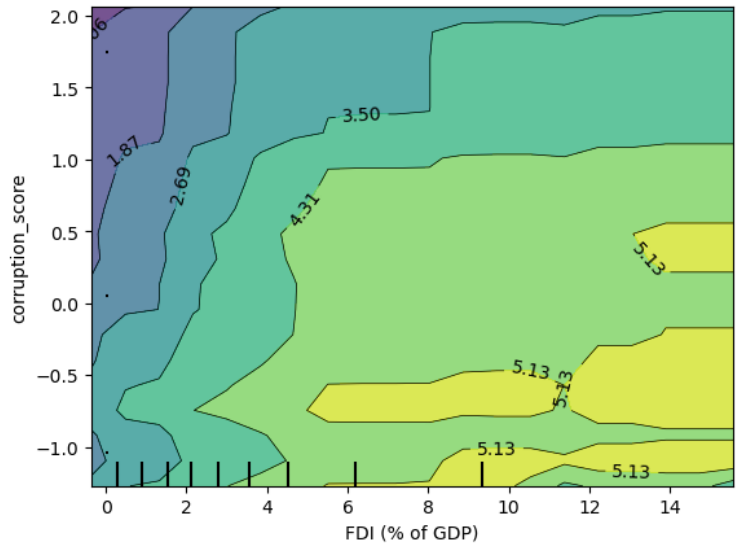

Figure 5: Partial dependence of GDP growth on FDI (% of GDP) and corruption score. The interaction effect is weaker than for political stability, but there is some indication that FDI is more effective in low-corruption environments.

To support and validate the SHAP-based insights, Figures 4 and 5 present partial dependence plots (PDPs), which show the average predicted GDP growth as FDI varies across different levels of governance scores. In Figure 4, the slope of the surface confirms that FDI has a steeper positive effect when political

stability is low, with the effect flattening in high-stability contexts. Figure 5 shows a similar surface for FDI and corruption. However, the contours are more uniform, suggesting only modest interaction. FDI remains generally growth-promoting, but its marginal effect does not vary as strongly with corruption levels.

4 Conclusions and Summary

This project investigated how the effectiveness of foreign direct investment (FDI) in promoting economic growth varies across countries with different levels of political stability and corruption. Using a combination of fixed-effects panel regression and interpretable machine learning (XGBoost with SHAP), we analyzed a panel dataset of macroeconomic and governance indicators spanning over two decades.

Across both modeling frameworks, FDI consistently emerged as a strong positive driver of GDP growth. However, the benefit of FDI was not uniform. A key insight from our analysis is that political stability moderates the relationship between FDI and economic growth: FDI appears to contribute more to growth in countries with lower political stability. This finding was supported both statistically (via a significant interaction term in the regression) and visually (via SHAP interaction and partial dependence plots). One possible explanation is that in politically unstable environments—where domestic investment may be risk-averse or constrained—FDI can serve as a substitute that drives capital formation, technology transfer, and infrastructure development.

In contrast, the effect of corruption on the FDI-growth relationship was weaker and less consistent. The interaction term was statistically insignificant in the regression, and SHAP visualizations revealed only a modest trend where FDI might be slightly more effective in less corrupt countries. This suggests that while corruption may shape the broader institutional environment, it does not systematically alter how FDI translates into growth within the observed data range.

These results offer policy-relevant implications. For politically fragile states, facilitating and attracting FDI—even in challenging conditions—may yield significant developmental returns. Meanwhile, for stable countries, the marginal benefit of additional FDI may be lower, indicating a need to focus on investment quality or domestic innovation. At the same time, the muted role of corruption in our findings suggests that its influence may lie in deterring FDI inflows, rather than in moderating FDI's effectiveness once present—a distinction worth exploring further.

There are several limitations worth noting. First, our dataset, though broad, may mask within-country heterogeneity and regional dynamics. Second, governance indicators are perception-based and may carry measurement error or temporal lag. Third, our models are descriptive rather than causal: while we capture associations and interactions, we do not establish causal mechanisms. Finally, while XGBoost improves pattern detection, its interpretations—though aided by SHAP—remain correlational and sensitive to feature distributions.

In sum, this study highlights the value of combining econometric and machine learning methods to uncover nuanced relationships in macroeconomic data. It offers evidence that context matters—FDI's growth-enhancing effect is not uniform, but shaped by a country's political landscape. Future work could extend this analysis by exploring the sectoral composition of FDI, entry modes, or firm-level outcomes, which may reveal more about how investment interacts with governance and institutions on the ground.